



Economic effects of automated health monitoring based on sensors versus visual observation for dairy cows in early lactation

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ABSTRACT

The objective of this study was to compare the cash flow of lactating Holstein cows randomized to health monitoring strategies that relied exclusively on automated health alerts or visual observation for selecting cows for clinical examination from 3 to 21 DIM. Data from a randomized controlled trial were used to estimate total cash flow per cow and per slot up to 100 DIM. We expected that greater milk yield in early lactation for the group monitored with automation would result in greater cash flow than for the group including visual observation despite greater health monitoring and treatment costs. Lactating Holstein cows ($n = 1,192$) fitted with a neck-attached automated rumination time and physical activity monitoring system (Sensehub, Merck Animal Health) and that had milk weights collected automatically at every milking (MM27BC, DeLaval) were randomly assigned to the visual observation (VO = 594) or automated health monitoring (AHM = 598) treatment. Cows were selected for clinical examination based on automated health alerts (i.e., health index score <86 arbitrary units, daily rumination time <250 min, and reduction in daily milk yield $>20\%$) in the AHM treatment and exclusively based on visual observation of clinical signs of disease in the VO treatment. Health, milk yield, and herd exit outcomes, and prices for inputs and outputs were collected through 100 DIM. Cash flow per cow and per slot (unit of space occupied by a cow at a dairy) including milk income over feed costs, health monitoring and management costs, treatment costs, and replacement costs were estimated and compared using deterministic analysis with linear models and stochastic analysis. Regardless of the method of estimation for cows that remained in the herd, cows that exited the herd, or for all cows combined, cash flow differences from the deterministic analysis generally favored the AHM treatment. Likewise, the stochastic simu-

lations supported the hypothesis that most of the time, the AHM treatment would result in positive cash flow compared with the VO treatment. Notably, the magnitude of the differences between treatment groups varied substantially based on the effects of treatments on herd performance outcomes, the herd exit dynamics, prices for influential inputs and outputs, and the methods used to estimate cash flow. Implementing a fresh cow health monitoring program that relied exclusively on automated monitoring system alerts to select cows for clinical examinations was economically beneficial compared with a program that relied exclusively on visual observation to select cows for clinical examinations.

Key words: health monitoring, automation, disease, dairy cow profitability

INTRODUCTION

Health disorders during the early postpartum period of lactating cows influence the profitability of dairy farms through reduced milk yield, increased risk of herd exit, reduced reproductive performance, and increased treatment and labor costs (Bareille et al., 2003; Hostens et al., 2012; Liang et al., 2017). To mitigate these negative effects of poor cow health on herd performance and cow well-being, monitoring programs are implemented to identify cows that require treatments or interventions (LeBlanc, 2010; Espadamala et al., 2016; Kerwin et al., 2022). Timely and appropriate treatments help cows recover from clinical signs, reduce the probability of developing concomitant disorders, return to normal milk production levels, and minimize downstream effects on performance.

Automated monitoring systems (AMS) that use behavioral, physiological, and performance parameters to identify cows potentially affected by health disorders can either replace or complement traditional health monitoring programs based on visual observation (VO) of clinical signs of disease, mandatory clinical examination, or combinations thereof. Different strategies for health monitoring programs using AMS have been evalu-

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The list of standard abbreviations for JDS is available at adsa.org/jds-abbreviations-26. Nonstandard abbreviations are available in the Notes.

ated and compared with traditional monitoring methods. Some experiments conducted at commercial farms suggested that health monitoring programs that relied on AMS alerts could be as effective as intensive monitoring strategies based on traditional methods to select cows for clinical examination (Silva et al., 2021; Perez et al., 2023). The results of the randomized controlled experiment presented in Rial et al. (2024) demonstrated that a health monitoring program designed to rely exclusively on AMS alerts was more effective than a nonintensive program that relied on VO for identifying Holstein cows with health disorders. Collectively, the few randomized controlled experiments conducted to evaluate health monitoring programs that relied either exclusively or almost exclusively on AMS alerts suggested the use of these technologies might be feasible for some commercial farms. However, none of these previous experiments evaluated the economics of the strategies incorporating AMS.

Therefore, the objective of this study was to determine the effects on cash flow of a health monitoring strategy that relied exclusively on AMS alerts to select cows for clinical examinations compared with a strategy that selected cows for clinical examinations by VO only. Specifically, we used data from the experiment reported in Rial et al. (2024) to estimate total cash flow per cow and per slot for a fixed period of 100 d postpartum. A secondary objective was to use stochastic Monte Carlo simulation to estimate cash flow differences between treatments for a range of input and output prices that could represent the market conditions in which dairy farms operate. We hypothesized that the improved performance observed for cows managed with AMS alerts during the randomized controlled experiment would result in greater cash flow despite increases in health monitoring and management costs associated with the use of automated monitoring.

MATERIALS AND METHODS

All procedures performed with cows were approved by the Animal Care and Use Committee of Cornell University (protocol no. 2017-0024).

Experimental Procedures

Details about cows, facilities, management, and additional experimental procedures are reported in Rial et al. (2024). Briefly, the study was conducted from April to October 2023 at a commercial dairy farm in Winsor, Colorado. The farm housed an average of 5,325 (range: 5,178 to 5,430) milking and 672 (range: 522 to 796) dry Holstein cows. Lactating cows were housed by parity (primiparous and multiparous apart) in 6-row freestall barns with concrete alleys and fed a TMR 2×/d with ad

libitum access to feed and water. Diets were formulated to meet or exceed the nutrient requirements of lactating dairy cows producing 42.0 kg/d milk with 4.1% fat and 3.1% true protein, as indicated by the Cornell Net Carbohydrate and Protein System (CNCPS v. 6.5.5). Daily milk yield (MY) per cow for the whole herd during the study period was 41.6 kg/d (range: 38.9 to 52.1 kg/d) and accumulated MY to 305 d for cows that calved during the study period was 11,308 kg (range: 4,395 to 16,810 kg). Cows were milked 3×/d at approximately 8-h intervals in a 90-stall rotary parlor. Cows had ad libitum access to water and a TMR delivered 3×/d and pushed 1×/h.

All cows enrolled in the experiment ($n = 1,225$) had a neck-attached behavior monitoring sensor tag of an AMS (Sensehub, Merck Animal Health) from at least 14 d before the expected date of parturition. After calving daily milk weights were collected automatically at every milking for individual cows using milk meters (MM27BC, DeLaval) at a 90-stall rotary parlor.

Briefly, the experiment followed a randomized controlled blocked design using parity (i.e., primiparous vs multiparous) and feeding trial treatment group during the close-up period (DCAD vs. mineral binder; not the objective of this experiment) as blocking factors. Multiparous cows were also stratified by previous lactation MY. At calving, cows were randomly assigned to a health monitoring program that used VO ($n = 594$) or a program that used automated health monitoring alerts (AHM; $n = 598$) to select cows for clinical examination from 3 to 21 DIM.

The following procedures were used to identify cows for clinical examination and diagnose clinical health disorders in each treatment:

VO Treatment

Cows were selected for clinical examination based exclusively on the observation of clinical signs of disease during a once-per-day walkthrough of pens ($n = 4$) where cows remained until ~21 DIM. Individual cows were inspected while in their pen after the first morning milking by a single member of the research team. Visual observation of all cows enrolled in the experiment was conducted by the first author (C.R.; veterinarian with 6 yr of experience and formal training in clinical examination and detection of health disorders in dairy cattle). Each cow housed in those pens underwent systematic observation. Visual observation of the cow front was used to evaluate general attitude (normal = alert, head and ears held up, eyes bright and no recession into the orbit vs. depressed = unresponsive, droopy ears, sunken eyes, lethargic), nasal discharge (0 = normal serous discharge, 1 = minimal cloudy discharge, 2 = cloudy and excessive discharge, 3 = copious, bilateral mucopurulent nasal

discharge), and abnormal breathing (shallow and rapid breathing) if present. Visual observation of the perineal area and udder was used to evaluate the presence of fetal membranes, vaginal discharge appearance (if present), manure consistency if the cow defecated (1 = watery, 2 = loose, 3 = thick porridge forming a pile of 2 to 3 cm, 4 = well-formed and stacks in rings, 5 = dry), and udder appearance if cows were standing (i.e., shape, color). All 4 legs were observed to determine inflammation of the hooves and signs of pain while walking if the cow was walking. A cow had to present at least one of the following to be selected for a clinical exam: depressed attitude; nasal discharge score >2 ; shallow or rapid breathing (or both); manure consistency scores of 1, 2, or 5; presence of fetal membranes; apparent signs of inflammation (i.e., redness, swelling) or pain in the udder; and abnormal vaginal discharge (red-brown, watery, fetid discharge). Cows selected for clinical examination were examined the next morning per the farm herd management procedures. Additional details of procedures are described in Rial et al. (2024).

AHM Treatment

Cows were selected for clinical examination based on alerts and data from the neck-attached sensor tag and MY. Specifically, cows underwent a clinical exam if they had a health index score <86 arbitrary units or daily rumination <250 min at any of the 2-h cycles of the 24-h period between 0530 and 0500 h of the following day. Cows were also selected for clinical examination if a reduction in MY $>20\%$ for the third milking session of the day (0500 h) compared with the rolling average of the third milking session of the 7 d before the day of interest was observed. Calculations started at 3 DIM so that data for at least 2 d before the day of calculation were available.

For both treatments, the list of cows for clinical evaluation was created daily at ~ 0530 h by a member of the research team (C.R.). Thus, every morning a list of cows selected for clinical examination was generated including (1) cows from the AHM group with health alerts generated up to the time the report was generated (including all alerts generated during the previous 24 h), (2) cows from the VO group identified by visual observation the previous day, and (3) cows previously diagnosed with a health disorder who were examined every other day until resolution of clinical signs of disease.

Cows underwent the procedures set forth by experimental treatments through 21 DIM or the day they left the fresh pen. From 22 to 100 DIM, cows from both treatment groups were sorted for clinical evaluation and examined using the same procedures (see details in the “Clinical Examination and Health Disorder Diagnosis” section).

Clinical Examination and Health Disorder Diagnosis

Clinical examination and health disorder case definitions are described in more detail in Rial et al. (2024). Briefly, clinical examinations were conducted by 3 technicians. Once cows on the list for examination were restrained in self-locking headgates and ready for examination, a technician in charge of identifying cows was located in the feed lane in front of cows for note-taking and evaluation of cow attitude and interest in feed. This person also indicated to the technician in charge of clinical exams (positioned behind cows) if a cow was eligible for a clinical exam or to receive an ongoing treatment. The third technician applied all treatments. The clinical examination procedure included the measurement of rectal temperature, evaluation of acetoacetic acid concentration in urine using the Ketostix test (Ketostix ketone strips, Bayer), and abdominal percussion to detect resonant areas as evidence of abnormal gas accumulation. Other examinations included rumen fill, manure characteristics, vaginal and uterine discharge, and udder abnormalities. Cows with pyrexia (rectal temperature $\geq 39.7^{\circ}\text{C}$), abnormal manure consistency (score 1, 2, or 5), Ketostix results different from negative, depressed attitude, and abnormal vaginal discharge (red-brown, watery, fetid discharge) during the clinical examination procedure had abdominal auscultation (assessment of internal body sounds using a stethoscope) conducted to examine primary ruminal movements. Thoracic auscultation was also conducted to evaluate lung sounds. In addition, cows with a Ketostix result different from negative had blood BHB concentrations measured using BHBCheck Plus strips (PortaCheck Inc., Moorestown, NJ).

The following criteria were used to define health disorders of interest: clinical ketosis = blood BHB concentration ≥ 1.2 mmol/L and at least 2 of the following clinical signs: lack of appetite, depressed attitude, decreased primary ruminal movements, and altered manure (too dry or too loose); displaced abomasum (DA) = abomasum in an abnormal position on the right or left side of the abdomen, detected by the presence of a metallic sound (i.e., “ping”); indigestion = either dry manure or diarrhea, lack of appetite, rumen, and intestinal stasis (determined by the presence of a dispersed metallic sound while simultaneously auscultating and percussion of either the right or left side of the abdomen) in the absence of DA and clinical ketosis with no signs of systemic illness; severe indigestion = either dry manure or diarrhea, lack of appetite, rumen, and intestinal stasis (determined by the presence of a dispersed metallic sound while simultaneously auscultating and percussion of either the right or left side of the abdomen) in the absence of DA and clinical ketosis with depressed attitude, pyrexia, or both. Other disorders of interest were defined as nonsevere metritis = watery,

red-brown, and fetid uterine discharge with or without pyrexia (rectal temperature $\geq 39.7^{\circ}\text{C}$ [103.5°F]); severe metritis = watery, red-brown, fetid uterine discharge with pyrexia and at least one of the following systemic signs of disease: depressed attitude, anorexia, dehydration, and decreased primary ruminal movements; nonsevere mastitis = visibly abnormal milk secretion (off-color, presence of clots, or blood) with or without signs of udder inflammation, such as hardness, redness, and increased udder skin temperature to the touch; severe mastitis = visibly abnormal milk secretion with or without signs of udder inflammation, as described previously, in addition to pyrexia and at least one of the systemic signs of disease described for severe cases of metritis; and pneumonia = rapid and shallow breathing, moist cough, nasal discharge, wheezing, depressed attitude, with or without pyrexia.

In addition, retained placenta was defined as failure to expel fetal membranes for up to 24 h after parturition and milk fever was defined as the inability to rise, or inability to rise and low body temperature, muscle tremors, and decreased ruminal motility observed up to 3 DIM.

Application of Treatments for Health Disorders

Doses and prices for each medication are presented in Table 1. Cows diagnosed with clinical ketosis received 300 mL of propylene glycol for 3 consecutive days. All cows diagnosed with DA underwent corrective surgery (left side abomasopexy) and were treated with ampicillin for 4 d and a unique dose of flunixin meglumine immediately after surgery. Cows with indigestion without signs of systemic illness received 3 magnesium hydroxide pills. Cows diagnosed with severe indigestion, in addition to the 3 magnesium hydroxide pills, received a drench containing 450 g of magnesium hydroxide in 20 L of water

and ampicillin trihydrate for 3 d. Cows diagnosed with nonsevere metritis received 2 doses of ceftiofur crystalline-free acid separated by 3 d. Cows with severe metritis were treated with ampicillin trihydrate for 3 d. Cows diagnosed with pneumonia received ampicillin trihydrate for 4 d and a unique dose of flunixin meglumine. Milk fever was treated with 500 mL of calcium gluconate, and another 500 mL of calcium gluconate was administered if cows were still unable to rise 12 h later. Cows diagnosed with retained placenta received 2 doses of ceftiofur crystalline-free acid separated by 3 d. For cows identified with nonsevere mastitis pathogen-specific treatments were applied. Based on farm protocols, milk cultures that resulted positive for *Streptococcus* spp. (non-*agalactiae*) or *Staphylococcus* spp. (non-*aureus*) received intramammary treatment with 1 tube of ceftiofur hydrochloride for 3 to 5 consecutive days depending on resolution of clinical signs. Cultures that were positive for *Klebsiella* spp., *Escherichia coli*, or had no growth observed, did not receive antimicrobial treatment. Cows for which the milk sample was positive for *Streptococcus agalactiae* or *Staphylococcus aureus* were relocated to a separate pen and sold. Finally, cows diagnosed with severe or toxic mastitis received hypertonic solution, ampicillin trihydrate for 3 d, a unique dose of flunixin meglumine, and intramammary antibiotic treatment following the criteria described for culture-based treatments.

Calculation of Economic Outcomes

Milk yield, herd exit (sales and deaths), and health performance outcomes were collected from DairyComp305 (Valley Agricultural Software) from parturition through 100 DIM to conduct a comprehensive analysis of cash flow for individual cows and slots occupied by cows over the period of interest. All revenues (milk sales, salvage

Table 1. Name, dose, and price of products used to treat cows diagnosed with clinical health disorders from 2 to 21 DIM for lactating Holstein cows randomized to treatments consisting of a health monitoring strategy that relied exclusively on automated health monitoring system alerts (AHM; n = 598) to select cows for clinical examination, or a strategy that selected cows for clinical examination based on visual observation (VO; n = 594) only from 3 to 21 DIM¹

Medication	Health disorder	Dose per application	Price, \$	Applications, n	Application interval, h	Milk withhold, h
Ceftiofur crystalline-free acid	Metritis or retained placenta	6.6 mg/kg BW	0.07/kg of BW	2	72	—
Ampicillin trihydrate	Metritis or pneumonia or DA	11.1 mg/kg BW	0.02/kg of BW	3 to 5	24	48
Flunixin meglumine	Metritis or pneumonia or DA	2.2 mg/kg BW	0.01/kg of BW	1	—	36
Calcium gluconate	Milk fever	500 mL/cow	5.5/500 mL	1 to 2	12	—
Propylene glycol	Clinical ketosis	300 mL/cow	3.4/300 mL	3	24	—
Magnesium hydroxide powder	Indigestion	450 g/cow	2.4/450 g	1	—	—
Magnesium hydroxide pills	Indigestion	3 pills/cow	1.5/3 pills	1	—	—
Hypertonic solution	Dehydration	2 L/cow	3.5/L	1	—	—
Ceftiofur hydrochloride IMM	Mastitis	1 tube/quarter	5.0/tube	3 to 5	24	72

¹DA = displaced abomasum; IMM = intramammary.

value of cows sold, and calf value for calves born from replacements) and expenses (feed costs, replacements costs, sensor costs, and labor costs) most closely affected by health monitoring and management programs and cow performance were calculated for 100 DIM or up to the day of herd exit for cows that were sold or died.

Based on these revenues and expenses, daily and accumulated cash flow per cow up to 100 DIM, and daily and accumulated cash flow per slot up to 100 DIM were calculated. A 100-d fixed period was chosen to allow expression of the potential long-term effects of treatments on the outcomes of interest. Cash flow per slot for 100 DIM was calculated to represent the conditions of commercial dairy farms that maintain constant herd size by replacing cows that exit the herd with a first-lactation cow. Therefore, we assumed that slots (i.e., unit of space at the dairy) initially occupied by a lactating cow enrolled in the experiment would be continuously occupied during a 100-d period starting on the day of parturition. All costs and revenues were accumulated for each day and the whole period up to 100 DIM. For cows that reached 100 DIM (i.e., cows that were not sold or died), all costs and revenues for the individual slot corresponded to the same cow. Conversely, for cows that left the herd before 100 DIM, a replacement cow that started the first lactation filled the slot the day after the replaced cow left the herd. The replacement was selected randomly from the pool of cows from the same treatment group. The replacement occupied the slot until the end of the 100-d period. A first-lactation replacement cow that left the herd before the end of the 100-d period was also replaced by another cow until the slot was fully occupied up to the end of the 100 d. All costs and revenues generated by the individual cow or the multiple cows that occupied a slot were accumulated up to the end of the 100-d period.

Total cash flow per cow or per slot was estimated using the following equation:

$$\text{cash flow} = \text{IOFC} - \text{replacement costs} - \text{health monitoring and management costs} - \text{treatment costs},$$

where IOFC = daily milk income over feed costs, replacement costs = cost of replacing cows that left the herd due to sale or death; health monitoring and management costs = costs of implementation of the health monitoring and management programs, and treatment costs = costs of treatments for cows diagnosed with health disorders.

Milk Income Over Feed Costs

Daily milk income over feed costs (IOFC) was calculated by subtracting feed costs (kg of DMI $\times$ DM costs

per kg) from milk revenue (kg of milk $\times$ milk price per kg) during lactation. Milk price was the average monthly price reported by the Dairy Market Watch (2023) report from April 2023 to October 2023. The weighted average accounting for milk class (class I to IV) price and usage was \$0.43/kg (\$19.59 per hundredweight). Milk from the hospital pen was fed to calves, and therefore, for the days that cows were in the hospital pen, the price used for milk was \$0.21/kg to represent the approximate costs of milk replacer.

Feed costs were set at \$0.18 per kg of DM for the fresh cow diet and \$0.22 per kg of DM for the lactating cow diet, based on estimated costs per kilogram of feed for the rations used to feed cows enrolled in the experiment.

FCM Production Estimation

To estimate daily DMI, FCM was calculated using daily MY obtained by individual stall milk meters (MM27BC, DeLaval) and monthly test milk fat percentage as follows:

$$\text{FCM (kg/d)} = 0.4 \times \text{DMW} + 0.15 \times \text{Fat \%} \times \text{DMW},$$

where DMW = daily milk weight (kg/d) and Fat % = fat percentage in milk reported in monthly tests. Because daily data for daily milk fat percentage were not available, the fat percentage value corresponding to a specific monthly test was repeated for the entire 30-d period corresponding to the test (e.g., the first monthly test value was used from 1 to 30 DIM).

BW Estimation

The Korver function (van Arendonk, 1985) was used to estimate daily BW for individual cows using

$$\text{BW (kg)} = A \times \{1 - [1 - (B/A)^{1/3}] \times \exp(-C \times \text{AGE})\}^3 - (P1/P2) \times \text{DIM} \times \exp(1 - \text{DIM}/P2) + P3^3 \times \text{DPC}^3,$$

where A = mature live weight (kg), B = birth weight (kg), C = growth rate, AGE = cow age in days, P1 = maximum decrease in live weight during lactation (kg), P2 = time during lactation to reach minimum live weight (DIM), P3 = pregnancy parameter, and DPC = day after conception minus 50 (0 for the first 50 d of gestation). The average mature live weight (703 kg) and birth weight (40 kg) was based on data retrieved from the farm. Parameters C, P1, P2, and P3 were 0.004, 30, 60, and 0.0187 for the first lactation, and 0.006, 50, 80, and 0.0187 for the second and third lactations, respectively (Kalantari et al., 2010).

DMI Estimation

Dry matter intake was estimated daily for individual cows using NRC (2001) equations. During lactation, DMI was estimated as a function of FCM, BW, and week of lactation, as follows:

$$\text{DMI (kg/d)} = 0.372 \times \text{FCM} + 0.0968 \times \text{BW}^{0.75} \\ \times \{1 - \exp[-0.192 \times (\text{DIM}/7 + 3.67)]\}.$$

Replacement Costs

The transaction costs of replacing cows that left the herd due to sale or death was equal to the value of a replacement heifer minus the salvage value of the replaced cow (\$0 if the cow left the herd due to death) and the value of the calf born from the replacement. The salvage value for cows sold was calculated based on the estimated BW at the time of sale multiplied by the average beef price.

To better represent conditions for farms that raise replacements versus those that buy replacements at an average market price, 2 separate replacement costs calculations were conducted. One calculation used the average market value for replacement heifers. This value was estimated at \$1,760 from the average price of replacement heifers reported by the USDA from April to July 2023 (USDA Economic Research Service, 2023). The approximate rearing costs of heifers from birth to parturition was used to estimate the rearing value replacement costs. The value used was \$2,355, which was the average cost to raise heifers in commercial dairy farms reported by Karszes and Hill (2020). The value of beef for cows sold was set at \$2.10/kg based on average beef price from April to July 2023 reported by the USDA National Agricultural Statistics Service (2023). Each cow that calved during the experiment had a record of the type of calf born, which was used to calculate the replacement cost. The value of a calf born from a replacement heifer was \$123 for Holstein females, \$301 for Holstein males, \$551 for cross-bred females, \$630 for cross-bred males, and \$0 for stillborn calves. A value of \$301 per calf was used for male–female twins assuming that females were freemartins. The value for each type of calf was set based on the average price for each category from April to July 2023 reported by the USDA National Dairy Comprehensive Report (USDA Economics, Statistics, and Market Information System, 2023).

Health Monitoring and Management Costs

For the VO group, the estimated costs of conducting the walkthrough of the fresh pen to select cows for clinical examination was estimated and assigned to individual cows. The average daily cost of conducting VO was esti-

mated at \$0.05 per cow per day based on a cost per hour of labor of \$19 and 30 min (average time required during the experiment reported in Rial et al., 2024) required for a single technician to complete the walkthrough of the pen and select cows for examination. The final daily cost per cow was estimated by dividing the total cost per day by an estimated average number of cows in the VO group in the fresh pen for the duration of the experiment (reported in the results section).

For the AHM group, the daily cost of the monitoring program was estimated and assigned to individual cows including only the cost of the automated monitoring system based on the neck-attached sensor tag. The cost of using daily MY alerts was assumed to be \$0 under the assumption that milk meters were not installed for health monitoring and there is no meter operating cost. The cost of using the neck-attached sensor tag per cow was estimated based on a fixed cost of \$2.58 per tag per month stipulated for farms with more than 2,000 active tags. The total cost per cow was then estimated by multiplying the daily cost to maintain a functional tag by the number of cow-days when a functional sensor was needed assuming each cow had a functional sensor on from 3 to 21 DIM.

To calculate labor costs for clinical examinations and treatments for both experimental treatment groups, the time spent in 3 types of procedures conducted after a cow was selected for clinical examination was estimated. The specific procedures which were the same for the VO and AHM groups were a basic clinical exam, a basic clinical exam plus complementary tests (i.e., BHB measurements, rumen auscultation, milk forestripping), and a basic exam plus complementary tests and application of treatments.

Based on records collected during the experiment, the estimated time spent per cow for these procedures was 1.5 min for a basic clinical examination, 2.5 min for a basic clinical exam plus complementary tests, and 7 min for a basic exam plus complementary tests and application of treatments.

The total daily cost per cow was then estimated by multiplying the time spent on each cow depending on the type of procedure that each cow underwent for each day by the cost of labor per minute assuming \$19/h of labor per technician.

Treatment Costs

The costs of treatments per dose or treatment unit for health disorders are specified in Table 1. Prices used for products were obtained from the Cornell University Ambulatory and Production Services veterinary product supply office. Each cow was assigned the cost per dose or unit of every treatment applied based on treatment

records collected during the experiment. The total treatment costs per cow per day were estimated by adding the costs of all treatments for individual cows for any given day. For example, all cows diagnosed with milk fever, regardless of BW, received 500 mL of calcium gluconate. Therefore, the price of this treatment was added every time that cows received a bottle of calcium. For antibiotics and anti-inflammatory drugs, a price per kilogram of BW for cows treated was calculated. To estimate the costs of each treatment, the estimated cost per kilogram of BW was multiplied by the estimated cow BW for that day.

Statistical Analysis

All statistical analyses were performed using SAS software (version 9.4, SAS Institute Inc., Cary, NC).

Descriptive statistics and frequency distributions for daily values for the parameters of interest were generated with PROC MEANS and PROC UNIVARIATE. The normality of continuous data was examined before analyses using graphical methods (histogram) and the Shapiro–Wilk test of normality generated by PROC UNIVARIATE.

Although treatments were applied from 3 to 21 DIM, all analyses included data from 2 DIM to have a baseline level for comparison.

Cash flow was analyzed separately for cows that remained in the herd and cows that exited the herd up to 100 DIM to account for the large difference in the distribution of cash flow for these subgroups of cows. Combining data from cows that remained and left the herd resulted in a highly skewed distribution with a normally distributed group of positive cash flow values and an extremely skewed distribution of negative values. This was the result of positive cash flow values for all cows that remained in the herd and negative cash flow values for all cows that left the herd. For the latter group, the data presented a skewed distribution to the left due to replacement costs outweighing all sources of revenue (i.e., IOFC and calf value). This was particularly obvious for cows that exited the herd due to death as the salvage value for these cows was zero. No data transformation method evaluated corrected for the major lack of normality of residuals and homoscedasticity of variance when data were analyzed by ANOVA. Therefore, separate analyses were conducted for cows that remained or exited the herd. For cows that remained in the herd and had no replacement costs, a single cash flow value was generated because the estimation of cash flow per cow was equivalent to the estimation per slot as all cows filled the slot (i.e., unit of space at the dairy) from the beginning of the experiment until 100 DIM. Conversely, for cows that exited the herd, 4 different cash flow estimates were generated to

account for the effect of the herd exit dynamics on the cash flow of dairy herds (i.e., analysis per slot versus per cow) and the effect of different replacement cow costs (i.e., rearing versus market cost of a replacement). Cash flow was estimated to 100 DIM per cow to evaluate the effect of treatments on individual cow cash flow and per slot filled to 100 DIM for every cow enrolled at the beginning of the experiment to better represent the effect of the herd exit dynamics on cash flow. The methods for estimation of cash flow described in Stangaferro et al. (2018, 2019) and Sitko et al. (2023) with modifications for the scenarios generated in the randomized controlled experiment of Rial et al. (2024) were used. Briefly, an individual slot (i.e., unit of space at the dairy) was the experimental unit and was aimed at representing the conditions of commercial dairy farms that maintain constant herd size throughout the year by replacing cows that exit the herd with a randomly chosen first-lactation cow from the same treatment group. Therefore, we assumed that slots initially occupied by a lactating cow enrolled in the experiment would be continuously occupied during 100 DIM starting on the day of calving. Cash flow per cow and per slot to 100 DIM was estimated using rearing costs or the market price of a replacement cow to represent these 2 scenarios of sourcing replacement cows.

Statistical analyses for the proportion of cows selected for clinical examination, proportion of cows identified with a clinical health disorder, and proportion of cows that received treatment and that left the herd were performed using logistic regression with the GLIMMIX procedure fitting a binomial distribution. For continuous outcomes (DMI, MY, days in the hospital pen, and incomes [milk income, IOFC, and cash flow for cows that remained and exited the herd]) ANOVA was performed with the MIXED procedure. Assumptions of normality and homoscedasticity of variance were evaluated with normal probability plots (normal Q-Q plot) and plots of residuals versus predicted values. For continuous outcomes not normally distributed (feed costs, replacement costs, treatment costs, health monitoring and management labor costs, and total expenses) a Mann–Whitney–Wilcoxon U-test was performed with the NPAR1WAY procedure because none of the data transformations tested corrected for the lack of normality of residuals. To account for the greater proportion of cows diagnosed and treated in the AHM group, treatment costs per health disorder of interest per cow enrolled in each treatment group (VO vs. AHM) were calculated by adding costs for each disorder for each treatment group divided by the total number of cows enrolled per treatment group.

Treatment groups were forced as a fixed effects in all models. Models for binary outcomes and normally distributed continuous outcomes were offered parity (primiparous vs. multiparous), calving-related events (yes =

at least one of the following: twins, abort, stillbirth, or dystocia, vs. no = none of the calving-related events observed), and close-up period feeding trial group (control vs. treatment) as covariates. For all outcomes, manual backward elimination of explanatory variables with $P > 0.10$ was adopted to select final models.

To estimate the total contribution of all cows enrolled in the experiment to total cash flow per cow and per slot, the weighted average of the differences between treatment groups was estimated based on the ratio of cows that stayed and left the herd for both treatment groups for each observation period. The average ratio of cows that stayed and left the herd for both treatment groups was calculated for each observation period to account for the actual proportion of cows that contributed to cash flow for each period.

Variables were considered significant if $P \leq 0.05$, whereas $0.05 < P \leq 0.10$ were considered a tendency. When appropriate, the least significant difference post hoc mean separation test was used to determine differences between least squares means. All proportions reported are least squares means, and the 95% confidence intervals were generated using the LSMEANS statement of GLIMMIX whereas quantitative outcomes were reported as least squares means $\pm$ standard error of the mean generated using the LSMEANS statement of the MIXED procedure.

Stochastic Analysis

Differences in cash flow between treatments (VO minus AHM) for each period of interest for all cows included in the study (i.e., cows that remained in the herd and cows that left the herd) under varying input and output values were estimated by stochastic Monte Carlo simulation models using @Risk software (version 7.5, Palisade Corp., Ithaca, NY). Simulations were run and recorded for 10,000 iterations with replacement. To build the models per cow and per slot, production outcomes and herd exit dynamics observed for the experiment were used as fixed inputs and are presented in Supplemental Tables S1 and S2 (see Notes), respectively. Thus, variation for every iteration of the simulation was introduced for milk price, feed costs, beef price (for culled cows), price of replacement heifers (market or rearing price), price of a newborn calves (Holstein female or male and cross-bred female or male), and health monitoring and treatment costs. Historical data were available for all inputs and outputs except health monitoring and treatment costs. Milk price (USDA Agricultural Marketing Service, 2023), beef price of cows sold (USDA National Agricultural Statistics Service, 2023), prices for replacement cows (USDA Economic Research Service, 2023), newborn Holstein calves (Pro-

gressive Dairyman, 2023), and feed costs for lactating cows (USDA Economic Research Service, 2023) were summarized from 2013 to 2023. The average, minimum, and maximum value for each parameter are presented in Supplemental Table S3 (see Notes). For these variables, distributions used in the simulations were fitted using the BestFit function of @Risk based on data available. This function selects the best-fitting distribution based on the lowest value for the Akaike information criterion. Stochasticity for the rest of the inputs (i.e., variables without historical data such as health monitoring costs [i.e., sensor cost + labor cost], treatment costs, and the price of cross-bred calves) was generated using a triangular distribution with the average price calculated as the most likely value and a 25% reduction or increment as minimum and maximum values, respectively. Values for the parameters used for each distribution are presented in Supplemental Table S4 (see Notes). All price variables were treated as independent in the stochastic simulations, which is the default in @Risk when no correlation structure is specified.

RESULTS

Twenty-one cows were removed from the experiment because they had no functional sensor tag either at calving ($n = 16$) or during the first 21 DIM ($n = 5$). In addition, 12 cows died before 3 DIM leaving a total of 1,192 cows in the experiment.

Effect of Treatments on Cows Examined, Identified with Health Disorders, and Treated

On average, 209 ± 14 cows for the AHM group and 205 ± 14 cows for the VO group were at risk of selection for clinical diagnosis per day and, 15.5 and 5.3 cows per d were included on the list for clinical examination for the AHM and VO group, respectively.

More ($P < 0.001$) cows in the AHM treatment than in the VO treatment underwent clinical examination, were diagnosed with ≥ 1 health disorder, and received treatments from 3 to 21 DIM (Table 2). More multiparous than primiparous cows ($P < 0.001$) and more cows with than without calving-related events ($P < 0.001$) underwent clinical examination.

Out of the cows examined, 66.2% and 45.8% were diagnosed with ≥ 1 health disorder from 3 to 21 DIM if in the VO and AHM group, respectively. More multiparous than primiparous cows ($P < 0.001$) and more cows with than without calving-related events ($P < 0.001$) were diagnosed with health disorders.

Average total treatment costs per cow enrolled in the experiment for each disorder of interest are presented in Figure 1. Overall, 62.7% (27/48) of the cows diagnosed

Table 2. Least squares means and 95% CI for the proportion of cows examined, cows that were identified with clinical health disorders, cows that received treatment, and the proportion of cows that left the herd (sold or dead) for lactating Holstein cows randomized to treatments consisting of a health monitoring strategy that relied exclusively on automated health monitoring system alerts (AHM; $n = 598$) to select cows for clinical examination, or a strategy that selected cows for clinical examination based on visual observation (VO; $n = 594$) only from 3 to 21 DIM

Item	Treatment, % [95% CI]		P-value
	VO ($n = 594$)	AHM ($n = 598$)	
Cows examined	24.2 [20.7, 28.1]	60.6 [56.3, 64.7]	<0.001
Cows diagnosed with clinical health disorders	22.3 [14.8, 32.2]	36.7 [26.6, 48.1]	<0.001
Cows treated	20.0 [13.2, 29.7]	32.7 [23.4, 43.9]	<0.001
Left herd to 21 DIM	8.3 [5.7, 12.0]	5.4 [3.6, 7.9]	0.09
Left herd from 22 to 100 DIM	8.6 [5.8, 12.4]	6.4 [4.49, 9.2]	0.21
Left herd to 100 DIM	16.7 [13.0, 21.4]	11.9 [9.2, 15.5]	0.05
Died to 21 DIM	1.6 [0.6, 4.0]	1.3 [0.5, 3.1]	0.64
Died from 22 to 100 DIM	1.9 [0.9, 5.1]	1.3 [0.7, 3.2]	0.29
Died to 100 DIM	2.8 [1.3, 6.1]	2.1 [0.9, 4.6]	0.40
Sold to 21 DIM	6.0 [3.8, 9.2]	3.2 [1.9, 5.5]	0.08
Sold from 22 to 100 DIM	7.1 [4.8, 10.9]	5.5 [3.8, 8.1]	0.37
Sold to 100 DIM	12.7 [9.4, 16.9]	9.1 [6.7, 12.3]	0.11

with mastitis did not receive antibiotic treatment because of the milk culture test outcome.

The average number of days that cows spent in the hospital pen did not differ ($P = 0.93$; VO: 6.3 ± 0.6 d; AHM: 6.3 ± 0.5 d) but more ($P = 0.02$) cows were placed in the hospital pen for ≥ 1 d for the AHM (15.8%) than the VO group (10.7%). Therefore, the cumulative number of days in the hospital pen was greater ($P < 0.001$) for the AHM (436 ± 3.4 d) than the VO group (277 ± 4.7 d). More multiparous than primiparous ($P < 0.001$) and more cows with than without calving-related events were in the hospital pen at least once ($P < 0.001$).

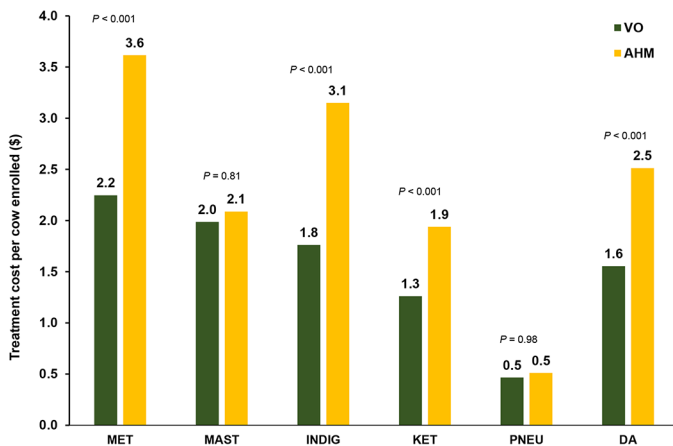


Figure 1. Treatment costs per cow for clinical health disorders of interest for lactating Holstein cows randomized to treatments consisting of a health monitoring strategy that relied exclusively on automated health monitoring system alerts (AHM; $n = 598$) to select cows for clinical examination or a strategy that selected cows for clinical examination based on visual observation (VO; $n = 594$) only from 3 to 21 DIM. MET = metritis; MAST = clinical mastitis; INDIG = indigestion; KET = clinical ketosis; PNEU = pneumonia; DA = displaced abomasum.

Herd Exit Dynamics

The total proportion of cows that left the herd (sold plus dead), were sold, or died is presented in Table 2. Overall, more cows with than without health disorders ($P < 0.001$) exited the herd for the periods analyzed. From 2 to 21 and from 22 to 100 DIM, more cows with than without health disorders ($P < 0.001$) died. For all periods analyzed, more multiparous than primiparous cows ($P = 0.01$) and more cows with than without health disorders ($P < 0.001$) died. Similarly, from 2 to 21 DIM, more multiparous than primiparous ($P = 0.004$) and more cows with than without health disorders ($P < 0.001$) were sold. From 22 to 100 and from 2 to 100 DIM, more cows with than without health disorders ($P < 0.001$) were sold.

Effect of Treatments on MY and Income, DMI, and Feed Costs

The effect of treatments on accumulated MY, milk income, total DMI, and feed costs per cow are presented in Table 3. For all periods analyzed, greater total MY, milk income, DMI, and total feed costs were observed for multiparous cows ($P < 0.001$), cows with no health disorders ($P < 0.001$), and cows with no calving-related events ($P < 0.001$).

The effects of treatment on accumulated MY, milk income, total DMI, and feed costs per slot are presented in Table 4. For all periods, greater total MY, milk income, DMI, and total feed costs were observed for multiparous cows ($P < 0.001$), cows with no health disorders ($P < 0.001$), and cows with no calving-related events ($P < 0.001$) at any time point analyzed.

Table 3. Least squares means $\pm$ SEM for total milk yield, milk income, total DMI, and feed costs per cow from 2 to 21 DIM, 22 to 100 DIM, and 2 to 100 DIM for lactating Holstein cows randomized to treatments consisting of a health monitoring strategy that relied exclusively on automated health monitoring system alerts (AHM; $n = 598$) to select cows for clinical examination, or a strategy that selected cows for clinical examination based on visual observation (VO; $n = 594$) only from 3 to 21 DIM

Item	Treatment, LSM $\pm$ SEM		P-value
	VO ($n = 594$)	AHM ($n = 598$)	
From 2 to 21 DIM			
Total milk, kg/cow	495 $\pm$ 10.7	523 $\pm$ 9.7	0.004
Total milk income, \$/cow	205 $\pm$ 4.7	217 $\pm$ 4.2	0.006
Total DMI, kg/cow	353 $\pm$ 4.5	358 $\pm$ 4.4	0.15
Total feed costs, \$/cow	63 $\pm$ 0.8	64 $\pm$ 0.8	0.14
From 22 to 100 DIM			
Total milk, kg/cow	3,094 $\pm$ 39.5	3,123 $\pm$ 37.4	0.41
Total milk income, \$/cow	1,329 $\pm$ 17.0	1,341 $\pm$ 16.1	0.42
Total DMI, kg/cow	1,876 $\pm$ 15.4	1,891 $\pm$ 14.5	0.27
Total feed costs, \$/cow	411 $\pm$ 3.4	415 $\pm$ 3.2	0.22
From 2 to 100 DIM			
Total milk, kg/cow	3,471 $\pm$ 43.7	3,523 $\pm$ 41.4	0.19
Total milk income, \$/cow	1,485 $\pm$ 18.8	1,507 $\pm$ 17.8	0.20
Total DMI, kg/cow	2,145 $\pm$ 17.2	2,167 $\pm$ 16.2	0.16
Total feed costs, \$/cow	456 $\pm$ 3.7	461 $\pm$ 3.5	0.14

Effect of Treatments on Accumulated Revenues and Expenses from 2 to 21 and 22 to 100 DIM

The effects of treatments on revenues and expenses per cow from 2 to 21 DIM and 22 to 100 DIM are presented in Table 5. For both periods, greater IOFC were observed for multiparous cows ($P < 0.001$), cows with no health disorders ($P < 0.001$), and cows with no calving-related events ($P < 0.001$).

The effect of treatments on revenues and expenses per slot from 2 to 21 DIM and 22 to 100 DIM is presented in Table 6. For both periods, greater IOFC were observed

for multiparous cows ($P < 0.001$), cows with no health disorders ($P < 0.001$), and cows with no calving-related events ($P < 0.001$).

The effects of treatments on accumulated revenues and expenses per cow and per slot from 2 to 100 DIM are presented in Table 7. Greater IOFC was observed for multiparous cows ($P < 0.001$), cows with no health disorders ($P < 0.001$), and cows with no calving-related events ($P < 0.001$). The total for all expenses combined per cow was greater for the AHM than the VO group when using market value ($P < 0.001$; AHM \$68.7 per cow and VO \$62.8 per cow) and rearing value replace-

Table 4. Least squares means $\pm$ SEM for total milk yield, milk income, total DMI, and feed costs from 2 to 21, 22 to 100, and 2 to 100 DIM per slot for lactating Holstein cows randomized to treatments consisting of a health monitoring strategy that relied exclusively on automated health monitoring system alerts (AHM; $n = 598$) to select cows for clinical examination, or a strategy that selected cows for clinical examination based on visual observation (VO; $n = 594$) only from 3 to 21 DIM

Item	Treatment, LSM $\pm$ SEM		P-value
	VO ($n = 594$)	AHM ($n = 598$)	
From 2 to 21 DIM			
Total milk, kg/slot	503 $\pm$ 10.6	534 $\pm$ 9.5	0.002
Total milk income, \$/slot	216 $\pm$ 4.5	227 $\pm$ 4.1	0.006
Total DMI, kg/slot	360 $\pm$ 4.3	366 $\pm$ 4.2	0.08
Total feed costs, \$/slot	65 $\pm$ 0.8	66 $\pm$ 0.7	0.08
From 22 to 100 DIM			
Total milk, kg/slot	3,433 $\pm$ 39.3	3,483 $\pm$ 37.2	0.16
Total milk income, \$/slot	1,408 $\pm$ 16.1	1,429 $\pm$ 15.2	0.17
Total DMI, kg/slot	2,145 $\pm$ 17.9	2,164 $\pm$ 16.9	0.26
Total feed costs, \$/slot	471 $\pm$ 3.9	475 $\pm$ 3.7	0.23
From 2 to 100 DIM			
Total milk, kg/slot	3,966 $\pm$ 44.5	4,035 $\pm$ 42.0	0.09
Total milk income, \$/slot	1,636 $\pm$ 18.4	1,663 $\pm$ 17.5	0.11
Total DMI, kg/slot	2,513 $\pm$ 20.7	2,538 $\pm$ 19.4	0.20
Total feed costs, \$/slot	537 $\pm$ 4.4	542 $\pm$ 4.2	0.17

Table 5. Least squares means $\pm$ SEM for total revenues and expenses accumulated per cow from 2 to 21 DIM and 22 to 100 DIM for lactating Holstein cows randomized to treatments consisting of a health monitoring strategy that relied exclusively on automated health monitoring system alerts (AHM; $n = 598$) to select cows for clinical examination, or a strategy that selected cows for clinical examination based on visual observation (VO; $n = 594$) only from 3 to 21 DIM

Item	Treatment, LSM $\pm$ SEM		P-value
	VO ($n = 594$)	AHM ($n = 598$)	
From 2 to 21 DIM, \$/cow			
Milk IOFC ¹	142 $\pm$ 4.1	153 $\pm$ 3.6	0.005
Market value replacement costs	30.4 $\pm$ 8.3	33.8 $\pm$ 9.1	0.61
Rearing value replacement costs	60.5 $\pm$ 12.4	65.5 $\pm$ 13.6	0.78
Health monitoring and management costs	2.2 $\pm$ 0.10	6.2 $\pm$ 0.10	<0.001
Treatment costs	5.8 $\pm$ 0.5	8.2 $\pm$ 0.6	<0.001
Calf value	12.3 $\pm$ 2.6	14.4 $\pm$ 4.4	0.57
From 22 to 100 DIM, ² \$/cow			
Milk IOFC	919 $\pm$ 14.1	927 $\pm$ 13.3	0.48
Market value replacement costs	25.6 $\pm$ 7.8	18.7 $\pm$ 6.2	0.92
Rearing value replacement costs	56.2 $\pm$ 11.9	48.1 $\pm$ 10.2	0.86
Calf value	15.6 $\pm$ 3.1	14.6 $\pm$ 3.0	0.68

¹Milk IOFC = milk income over feed costs.

²Health monitoring and management and treatment costs not estimated from 22 to 100 DIM.

ment costs ($P < 0.001$; AHM \$125.5 per cow and VO \$121.9 per cow).

When calculated per slot, greater IOFC per slot was observed for multiparous cows ($P < 0.001$), cows with no health disorders ($P < 0.001$), and cows with no calving-related events ($P < 0.001$). The total for all expenses combined per slot was greater for the AHM than the VO group when using market value ($P < 0.001$; AHM \$83.8 per cow and VO \$78.1 per cow) and rearing value replacement costs ($P < 0.001$; AHM \$142.3 per cow and VO \$140.9 per cow). Total expenses were greater for cows with health disorders ($P < 0.001$) and cows with calving-related events ($P < 0.001$).

Effect of Treatments on Accumulated Cash Flow per Cow and per Slot for All Periods

The effects of treatments on accumulated cash flow per cow for all periods are presented in Table 8. For all periods, greater cash flow was observed for multiparous cows ($P < 0.001$), cows with no health disorders ($P < 0.001$), and cows with no calving-related events ($P < 0.001$).

The effect of treatments on accumulated cash flow per slot for all periods for cows that left the herd is presented in Table 9. Note that cash flow per slot for cows that remained in the herd is the same as cash flow estimated per

Table 6. Least squares means $\pm$ SEM for total revenue and expense sources accumulated from 2 to 21 DIM and 22 to 100 DIM per slot enrolled for lactating Holstein cows randomized to treatments consisting of a health monitoring strategy that relied exclusively on automated health monitoring system alerts (AHM; $n = 598$) to select cows for clinical examination, or a strategy that selected cows for clinical examination based on visual observation (VO; $n = 594$) only from 3 to 21 DIM

Item	Treatment, LSM $\pm$ SEM		P-value
	VO ($n = 594$)	AHM ($n = 598$)	
From 2 to 21 DIM, \$/slot			
Milk IOFC ¹	144 $\pm$ 4.0	156 $\pm$ 3.6	0.002
Market value replacement costs	32.3 $\pm$ 8.5	43.7 $\pm$ 11.2	0.60
Rearing value replacement costs	61.3 $\pm$ 12.5	69.4 $\pm$ 14.6	0.79
Health monitoring and management costs	2.3 $\pm$ 0.1	6.4 $\pm$ 0.3	<0.001
Treatment costs	5.9 $\pm$ 0.6	8.4 $\pm$ 0.7	<0.001
Calf value	12.9 $\pm$ 2.6	15.1 $\pm$ 3.6	0.78
From 22 to 100 DIM, ² \$/slot			
Milk IOFC	1,029 $\pm$ 11.8	1,046 $\pm$ 10.8	0.21
Market value replacement costs	37.2 $\pm$ 9.2	24.6 $\pm$ 6.8	0.67
Rearing value replacement costs	72.0 $\pm$ 12.8	57.4 $\pm$ 10.9	0.51
Calf value	17.3 $\pm$ 3.2	15.8 $\pm$ 3.1	0.61

¹Milk IOFC = milk income over feed costs.

²Health monitoring and management and treatment costs not estimated from 22 to 100 DIM.

Table 7. Least squares means $\pm$ SEM for total revenue and expense sources accumulated from 2 to 100 DIM per cow and per slot enrolled for lactating Holstein cows randomized to treatments consisting of a health monitoring strategy that relied exclusively on automated health monitoring system alerts (AHM; $n = 598$) to select cows for clinical examination, or a strategy that selected cows for clinical examination based on visual observation (VO; $n = 594$) only from 3 to 21 DIM

Item	Treatment, LSM $\pm$ SEM		P-value
	VO ($n = 594$)	AHM ($n = 598$)	
From 2 to 100 DIM, \$/cow			
Milk IOFC ¹	1,029 $\pm$ 15.7	1,047 $\pm$ 15.7	0.22
Market value replacement costs	54.7 $\pm$ 10.9	54.5 $\pm$ 10.7	0.67
Rearing value replacement costs	113.8 $\pm$ 16.5	111.3 $\pm$ 16.4	0.75
Health monitoring and management costs	2.2 $\pm$ 0.10	6.1 $\pm$ 0.10	<0.001
Treatment costs	5.9 $\pm$ 0.5	8.1 $\pm$ 0.6	<0.001
Calf value	27.9 $\pm$ 3.9	29.0 $\pm$ 5.3	0.49
From 2 to 100 DIM, \$/slot			
Milk IOFC	1,171 $\pm$ 16.3	1,194 $\pm$ 15.4	0.12
Market value replacement costs	69.5 $\pm$ 13.5	68.3 $\pm$ 13.3	0.90
Rearing value replacement costs	132.3 $\pm$ 18.2	126.8 $\pm$ 18.0	0.56
Health monitoring and management costs	2.4 $\pm$ 0.1	6.8 $\pm$ 0.2	<0.001
Treatment costs	6.2 $\pm$ 0.6	8.7 $\pm$ 0.7	<0.001
Calf value	30.3 $\pm$ 4.2	31.0 $\pm$ 4.7	0.64

¹Milk IOFC = milk income over feed costs.

cow, and data are presented in Table 8. For all periods, greater cash flow was observed for multiparous cows ($P < 0.001$), cows with no health disorders ($P < 0.001$), and cows with no calving-related events ($P < 0.001$).

Weighted Average of Cash Flow for All Cows Combined

The weighted average of the differences between treatment groups for cash flow for all cows enrolled in the

experiment for each observation period is presented in Table 10. The weighted average was estimated based on the ratio of cows that stayed and left the herd for both treatment groups (Table 2).

Stochastic Analysis

Negative values indicate greater cash flow for the AHM than the VO treatment group because differences were estimated by subtracting the cash flow of the AHM from

Table 8. Least squares means $\pm$ SEM for total cash flow accumulated from 2 to 21, 22 to 100, and 2 to 100 DIM per cow for lactating Holstein cows randomized to treatments consisting of a health monitoring strategy that relied exclusively on automated health monitoring system alerts (AHM; $n = 598$) to select cows for clinical examination, or a strategy that selected cows for clinical examination based on visual observation (VO; $n = 594$) only from 3 to 21 DIM

Item	Treatment, LSM $\pm$ SEM		P-value
	VO	AHM	
Cash flow for cows that remained in the herd, \$/cow			
From 2 to 21 DIM	127 $\pm$ 4.3 $n = 564$	139 $\pm$ 3.6 $n = 570$	0.007
From 22 to 100 DIM	1,042 $\pm$ 15.0 $n = 526$	1,048 $\pm$ 14.1 $n = 536$	0.64
From 2 to 100 DIM	1,179 $\pm$ 17.4 $n = 526$	1,189 $\pm$ 16.3 $n = 536$	0.48
Cash flow for cows that exited the herd, \$/cow			
From 2 to 21 DIM	$n = 30$	$n = 28$	
Cash flow (replacement market value)	-581 $\pm$ 129.2	-767 $\pm$ 134.3	0.32
Cash flow (replacement rearing value)	-1,142 $\pm$ 129.3	-1,267 $\pm$ 144.4	0.49
From 22 to 100 DIM	$n = 38$	$n = 34$	
Cash flow (replacement market value)	-47 $\pm$ 142.6	-39 $\pm$ 140.2	0.96
Cash flow (replacement rearing value)	-642 $\pm$ 142.6	-635 $\pm$ 140.2	0.84
From 2 to 100 DIM	$n = 68$	$n = 62$	
Cash flow (replacement market value)	-271 $\pm$ 101.9	-247 $\pm$ 102.4	0.86
Cash flow (replacement rearing value)	-866 $\pm$ 101.9	-843 $\pm$ 102.4	0.78

Table 9. Least squares means $\pm$ SEM for total cash flow accumulated from 2 to 21, 22 to 100, and 2 to 100 DIM per slot for cows that remained and exited the herd (died or sold) for lactating Holstein cows randomized to treatments consisting of a health monitoring strategy that relied exclusively on automated health monitoring system alerts (AHM; $n = 598$) to select cows for clinical examination, or a strategy that selected cows for clinical examination based on visual observation (VO; $n = 594$) only from 3 to 21 DIM

Item	Treatment, LSM $\pm$ SEM		P-value
	VO	AHM	
Cash flow for cows that remained in the herd, ¹ \$/slot			
Cash flow for cows that exited the herd, \$/slot			
From 2 to 21 DIM	$n = 30$	$n = 28$	
Cash flow (replacement market value)	-503 ± 150.1	-721 ± 167.6	0.31
Cash flow (replacement rearing value)	$-1,087 \pm 139.0$	$-1,266 \pm 155.3$	0.37
From 22 to 100 DIM	$n = 38$	$n = 34$	
Cash flow (replacement market value)	217 ± 123.4	359 ± 121.3	0.35
Cash flow (replacement rearing value)	-367 ± 120.9	-240 ± 118.9	0.39
From 2 to 100 DIM	$n = 68$	$n = 62$	
Cash flow (replacement market value)	249 ± 102.9	324 ± 103.4	0.59
Cash flow (replacement rearing value)	-295 ± 94.1	-234 ± 94.4	0.64

¹Values of cash flow per slot are the same as the values of cash flow per cow presented in Table 8.

the VO group. Summary statistics and the percentage of time that the cash flow difference favored the AHM treatment are presented in Table 11. For all scenarios and periods, the mean and median value of the difference in cash flow benefited the AHM treatment.

For the analysis of cash flow per cow from 2 to 21 DIM, milk price and replacement costs were the main contributors to the total variance for cash flow per cow. These inputs explained 28% and 29% of the total variation in cash flow differences, respectively, when using market price, and 28% and 29% of the total variation when using rearing costs for replacements. Greater milk and replacement costs prices favored the AHM treatment. For the analysis of cash flow per slot from 2 to 21 DIM, milk and

feed prices were the main contributors to the total variance. Milk and feed prices explained 38% and 26% of the total variation in cash flow differences, respectively, when using heifer market prices, and 36% and 27% of the variation when rearing replacement prices were used. Greater milk prices favored the AHM treatment whereas greater feed prices favored the VO treatment.

Similar than for the previous period, differences in cash flow per cow from 22 to 100 DIM were affected primarily by variation in milk prices and replacement costs. These inputs explained 40% and 42% of the total variation in cash flow differences, respectively, when using market price, and 39% and 44% of the total variation when using rearing costs for replacements. For

Table 10. Weighted average for the differences in cash flow between treatments for all cows enrolled in the experiment based on the ratio of cows that stayed and left the herd¹

Type of cash flow estimation for all cows combined			
Per cow		Per slot	
Market value replacement	Weighted average cash flow difference, ² \$/cow	Market value replacement	Weighted average cash flow difference, ² \$/slot
2 to 21 DIM	2.4	2 to 21 DIM	0.8
22 to 100 DIM	6.1	22 to 100 DIM	15.1
2 to 100 DIM	11.4	2 to 100 DIM	17.3
Rearing value replacement		Rearing value replacement	
2 to 21 DIM	5.3	2 to 21 DIM	2.7
22 to 100 DIM	6.1	22 to 100 DIM	14.1
2 to 100 DIM	11.3	2 to 100 DIM	15.7

¹Lactating Holstein cows were randomized to treatments consisting of a health monitoring strategy that relied exclusively on automated health monitoring system alerts (AHM; $n = 598$) to select cows for clinical examination or a strategy that selected cows for clinical examination based on visual observation (VO; $n = 594$) only from 3 to 21 DIM.

²Positive values indicate greater cash flow for the AHM than for the VO observation group.

Table 11. Summary statistics from a stochastic analysis that compared cash flow differences for lactating Holstein cows randomized to treatments consisting of a health monitoring strategy that relied exclusively on automated health monitoring system alerts (AHM; n = 598) to select cows for clinical examination, or a strategy that selected cows for clinical examination based on visual observation (VO; n = 594) only from 3 to 21 DIM¹

Item	Mean	SD	Median	Minimum	Maximum	Scenarios AHM > VO, %
Per cow (market value replacement)						
2 to 21 DIM	-4.3	3.4	-4.2	-21.3	11.2	80
22 to 100 DIM	-19.2	5.7	-18.6	-46.0	2.2	100
2 to 100 DIM	-7.9	3.1	-7.7	-33.7	11.8	90
Per cow (rearing value replacement)						
2 to 21 DIM	-9.8	3.7	-9.8	-28.0	5.2	99
22 to 100 DIM	-20.5	3.3	-20.4	-35.1	-7.3	100
2 to 100 DIM	-13.7	5.3	-13.5	-41.8	9.3	100
Per slot (market value replacement)						
2 to 21 DIM	-4.2	3.4	-4.1	-17.8	6.9	80
22 to 100 DIM	-19.2	5.7	-18.6	-47.0	2.2	100
2 to 100 DIM	-9.9	6.4	-9.6	-40.3	11.9	95
Per slot (rearing value replacement)						
2 to 21 DIM	-3.1	3.6	-3.1	-19.9	11.2	80
22 to 100 DIM	-32.0	5.9	-31.9	-59.5	-8.0	100
2 to 100 DIM	-16.9	6.4	-16.5	-55.6	12.2	99

¹Values generated by subtracting the cash flow of cows in the AHM from cows in the VO treatment.

the analysis of cash flow per slot from 22 to 100 DIM, milk price explained 36% and replacement costs 42% of the total variation in cash flow differences when using heifer market prices. For estimations using heifer rearing costs milk prices explained 37% and replacement costs 42% of the variation. In all cases, greater milk prices and replacement heifer costs favored the AHM treatment.

Finally, regarding the differences in cash flow per cow from 2 to 100 DIM (Table 11 and Figure 2), milk price and replacement costs were also the main contributors to the total variance. These inputs explained 42% and 29% of the total variation in cash flow differences, respectively, when using heifer market prices, and 47% and 17% of the total variation when using heifer rearing costs. The same inputs were the main contributors to variability in cash flow differences per slot. Milk price explained 56% and replacement costs 17% of the total variation in cash flow differences when using heifer market prices. For estimations using rearing heifer rearing costs milk prices explained 55% and replacement costs 19% of the variation. In all cases, greater milk prices and replacement costs favored the AHM treatment.

DISCUSSION

In the current study we hypothesized that cows managed with a health monitoring strategy that relied exclusively on AMS alerts to select cows for clinical examination would be more profitable than a program that used visual observation alone because greater MY would offset and outweigh greater health monitoring costs, clinical exam costs, and more cow-days in the hospital pen due to

the increased proportion of cows diagnosed with health disorders (Rial et al., 2024).

Cash flow was estimated and analyzed in multiple ways because of the effect of numerous complex factors on cow profitability. The estimation of cash flow per cow and per slot to 100 DIM captured longer-term effects of treatments (i.e., longer than the first 3 wk of lactation) rather than only when the treatments were applied. As the analysis of cash flow per cow only partially accounted for the herd replacement dynamics, the analysis per slot was used to fully account for the effects of cow herd exit to 100 DIM (i.e., slots were occupied by replacements). Although more representative of conditions of constant herd size (Stangaferro et al., 2018; Sitko et al., 2023), the analysis per slot did not account for the effects of treatments on cash flow past 100 DIM. Only analyses of cash flow per slot that cover the approximate lifetime of cows can truly quantify the effects of the treatments on a dairy herd. Due to the extreme negative values for cows that were sold or died, cash flow was also estimated separately for cows that remained versus those that exited the herd. Finally, cash flow was estimated using the market value versus the rearing costs of replacements to represent the large cash flow variation caused by these common approaches to sourcing replacement cows (Sitko et al., 2023).

Regardless of the method of estimation, cash flow differences from the deterministic analyses always favored the AHM treatment. For cows that remained in the herd, differences ranged from \$6 to \$12 in favor of the AHM treatment but were only statistically significant for the period from 2 to 21 DIM. However, for cows that left the herd cash flow was in favor of the AHM treatment

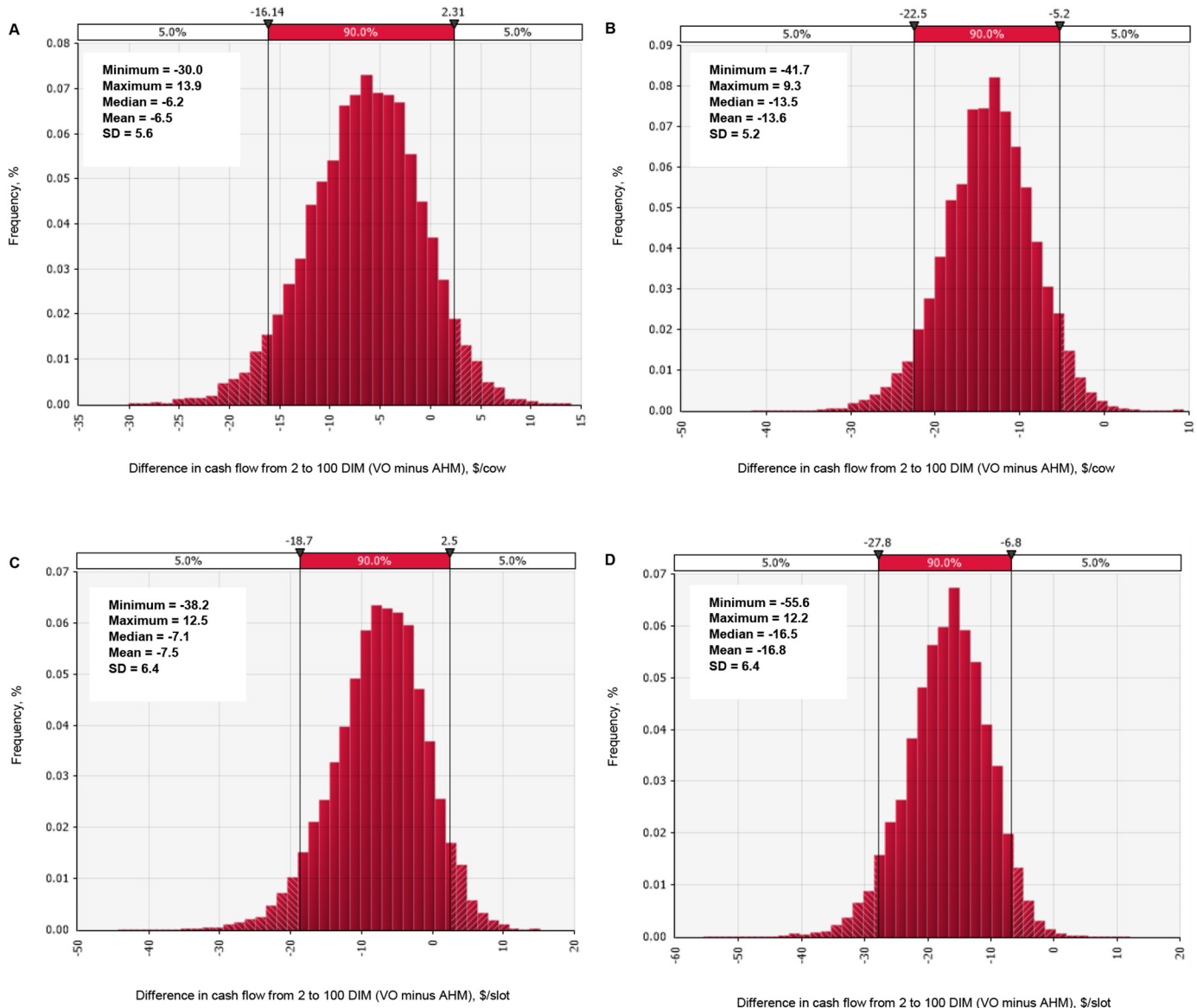


Figure 2. Stochastic analyses for the comparison cash flow from 2 to 100 DIM for lactating Holstein cows randomized to treatments consisting of a health monitoring strategy that relied exclusively on automated health monitoring system alerts (AHM; $n = 598$) to select cows for clinical examination or a strategy that selected cows for clinical examination based on visual observation (VO; $n = 594$) only from 3 to 21 DIM. (A) Differences in cash flow per cow using the market value of replacement, (B) per cow using the rearing value of replacement, (C) per slot using the market value of replacement, and (D) per slot using the rearing value of replacement.

for the 22 to 100 and the 2 to 100 DIM period but not for the first 21 DIM. Finally, despite large variability in the magnitude of the differences, the weighted averages for cash flow using the ratio of cows that remained and left the herd under all possible scenarios favored the AHM treatment from as low as \$2.4 per cow or \$0.8 per slot to as much as \$11.4 per cow and \$17.3 per slot. Thus, implementing a fresh cow health monitoring program that relies exclusively on AMS alerts to select cows for clinical examinations could be economically beneficial compared with a program that relies exclusively on

visual observation to select cows for clinical examinations. The magnitude of the benefit, however, could vary substantially based on the effects of these cow health monitoring strategies on herd performance, the herd exit dynamics, input and output prices, and the methods used to estimate cash flow.

Driven by greater IOFC from 2 to 21 DIM for cows that remained in the herd, the AHM treatment had greater accumulated cash flow to 21 DIM, supporting our hypothesis that the AHM treatment would have greater cash flow despite increased costs for labor, sensor use,

and treatments. Thus, the health disorder detection and intervention dynamics generated by the AHM treatment improved health outcomes and minimized performance losses to an extent that generated larger gains in income than losses due to increased expenses. The main driver of greater cash flow for the AHM group was increased IOFC across the first 100 d after calving. During the first 21 DIM when the treatments were implemented, the AHM group generated more IOFC per cow and per slot than the VO group. Although less influential in the total cash flow accumulated to 100 DIM, greater IOFC for cows in the AHM group during the first 21 DIM was the result of more MY for cows diagnosed with health disorders. These data suggested a direct short-term positive effect of improved cow health on profitability. Prompt identification and treatment, and more cows treated were expected to contribute to faster recovery and return to normal behavior and productivity. This is also supported by the improved rumination time observed for cows in the AHM group in the first 21 DIM reported in Rial et al. (2024). Conversely, the greater IOFC (not statistically significant) observed after 21 DIM for the AHM group was driven by the group of cows not diagnosed with clinical health disorders. For these cows, differences in milk IOFC became evident after 3 wk postpartum and differed significantly from cows in the VO group toward the end of the 100 DIM period (data not shown). From 12 to 14 wk after calving, differences in IOFC were in the range of \$0.002 to \$0.57 per cow per day and \$0.002 to \$0.44 d per slot per day. Unlike for the first 21 DIM when greater milk production for the AHM group was likely the result of short-term improvements on health in more cows diagnosed with health disorders, the +29 and +50 kg of milk per cow, and slot and +\$8 and +\$17 in IOFC per cow and slot (not statistically significant) observed from 22 to 100 DIM was likely the result of the long-term effects of poor health in early lactation on milk production on cows in the VO treatment. We speculate that some cows in the VO group affected by health disorders within the first 21 DIM were not identified and therefore did not benefit from therapeutic treatments. As these cows remained in the herd, their milk production was likely negatively affected by the health disorders experienced in early lactation. Long-term negative effects of early-lactation health disorders on milk production have been extensively documented (Ribeiro et al., 2016; Carvalho et al., 2019; Probo et al., 2018; Pinedo et al., 2020). Thus, compared with less effective strategies to select cows for clinical examination such as use of VO alone, the use of automated monitoring to identify cows for clinical examination could be economically beneficial not only because of improved health in the short term, but also because of a lesser influence of poor health on cow productivity in the long term. Overall, these findings, which

did not cover an entire lactation, provide evidence of the economic benefits of sensor-based health monitoring and emphasize the importance of accounting for both short- and long-term effects of integrating these technologies into herd management.

The relationships between dairy herd economics and the amount of effort and resources used to implement health monitoring and management programs and treatments are simpler than associations with milk production efficiency and replacement costs (Lima et al., 2019; Pérez-Báez et al., 2020). Indeed, results consistently supported the hypothesis that the AHM treatment would result in greater costs of implementation reducing the value of greater IOFC. The health monitoring and management program including automated alerts was more expensive to implement not only because of the cost of using the neck-attached AMS but also increased labor costs. Health monitoring and management costs were more than double and treatment costs were ~40% greater for the AHM than the VO group. Although this is relevant from an input cost perspective, the relative contribution to overall cash flow (i.e., <1% of total cash flow to 100 DIM) and the magnitude of the differences between treatments for these cost items were substantially smaller than those of IOFC. Ultimately, gains in IOFC over the 100-d period analyzed fully offset the greater cost of implementing the health monitoring program during the first 3 wk of lactation.

During the first 21 DIM, the AHM treatment was more expensive mostly because more cows were selected for clinical examination, identified with health disorders, and received treatments. In line with observations for the proportion of cows diagnosed with different health disorders, the increase in treatment costs was observed primarily for metritis and for metabolic-digestive disorders (clinical ketosis, indigestion, and DA). As reported in Rial et al. (2024), cows in the AHM group were at greater risk of being identified with metritis, clinical ketosis, and indigestion, which explained the greater treatment costs. Although there was only a statistical tendency for more cows to be diagnosed with DA in the AHM than in the VO group, the greater costs of DA treatment for the AHM group were likely due to the high cost of individual treatments for DA.

Driven by the negative cash cost of replacing most cows that exit herds, small differences in the replacement dynamics and the ratio of cows that leave due to sale or death can alter the direction and magnitude of estimated differences in cash flow (Stangaferro et al., 2018, 2019; Sitko et al., 2023). This is exacerbated when cash flow is accumulated for relatively short periods of time after calving because the cash cost of replacing cows, especially for dead cows, is disproportionately high (i.e., dead cows have no salvage value) compared with rev-

venues such as IOFC and calf value. Indeed, the reduction in the proportion of cows that left the herd for the AHM group did not translate into lower replacement costs, and from 2 to 21 DIM, the accumulated cash flow per cow or per slot for cows that left the herd was approximately +\$100 to \$200 more negative for the AHM treatment than for the VO treatment. This may be attributed to the few cows that exited the herd during this short study period and key drivers of the association between herd exit dynamics and cash flow including the ratio of cows that leave due to death or sale, and the parity distribution of cows that leave. In this case, out of all cows that left within treatment, more cows left due to death for the AHM (46%) than the VO (33%) treatment and more of the cows that left were primiparous for the AHM (64%) than the VO (53%) treatment. This caused a difference in salvage value (data not shown) that explained most of the differences in cash flow to 21 DIM for cows that exited the herd. Conversely, for the other periods, the cash flow differences balanced out to the point that the AHM treatment was favored. This can be explained by the smaller magnitude of replacement costs and the larger effect of IOFC on cash flow evaluated over longer periods of time. The consequences of the different magnitudes of the herd exit dynamics on cash flow for short versus long periods of time were evident. We observed a substantial reduction of the economic benefit of the AHM treatment when the cash flow weighted average was estimated to account for cows that stayed and left the herd. In contrast with the longer periods of estimation (22 to 100 and 2 to 100 DIM) for which cash flow did not change substantially, for the 2 to 21 DIM period the weighted average cash flow for cows that stayed and left the herd substantially reduced the benefit of the AHM treatment. The difference in favor of the AHM treatment when only cows that stayed were accounted for (i.e., \$12) was as low as \$0.8 per slot with market value for replacements to a maximum of \$5.3 per cow including rearing values for replacements. These results emphasized the complexity of the effects of the herd exit on cash flow estimations in studies comparing management strategies that alter health monitoring and management and demonstrated the value of monitoring consequences of treatments on cash flow in the medium to long term.

The sensitivity analysis including stochastic simulations was conducted to better understand the effects of treatments on cash flow under a wider range of farm management and market conditions (McArt and Oetzel, 2015; Liang et al., 2017; Stangaferro et al., 2018). In agreement with the deterministic analysis, the stochastic simulations supported the hypothesis that most of the time, the AHM treatment would result in improved cash flow. The AHM treatment had a positive difference in cash flow from 80% to 100% of the time. Moreover,

under the worst scenarios for the AHM treatment the economic loss would be much smaller than the possible gains under the most favorable scenarios. For example, the largest loss for the AHM treatment for the 2 to 100 DIM period would be \$12.2 per slot, whereas the largest gain would be \$55.6 per slot when using rearing costs for replacements. Despite a few exceptions, the mean differences between treatments from the stochastic analysis were within a relatively close range than those observed from the weighted averages for all cows in the deterministic analysis. Of note, the relatively lower frequency of positive cash flow differences per slot in favor of the AHM for the analysis from 2 to 21 DIM was because of randomly selecting more replacement cows with low cash flow. This issue not only serves as an example of the plausible substantial effects of the herd exit dynamics on cash flow estimations including replacement costs, but also of how random chance can affect results when including low incidence events such herd exit in early lactation.

Collectively, results from the stochastic analysis suggested that when accounting for biological and economic variability, the AHM strategy consistently outperformed the VO treatment. This likely reflected the compounded benefits of early disease detection and intervention, as well as a herd exit dynamics that resulted in having more higher-performing cows later in lactation. The use of probabilistic modeling demonstrated the robustness of the advantage of using automation to identify cows for clinical examinations and underscored the importance of evaluating management strategies under different pricing scenarios.

The most notable study limitations were the use of a single commercial farm, which limits the external validity of results, issues with internal validity owing to the biases of personnel who conducted the on-farm work, and covering only the first 100 d of lactation. Results may be more applicable to large commercial herds that select cows for clinical examination and conduct clinical exams once per day and strictly follow standard operating procedures for cow observations, clinical examination, and treatment decision-making. Not including data for the entire lactation excluded potential additional differences in cash flow. There were other pitfalls inherent to the design of the randomized controlled trial including the design of the experimental treatments and the specific features and intrinsic performance of the AHM systems used. Finally, including items in the cash flow estimations for which there were no statistically significant differences between treatments (e.g., replacement costs) could have unnecessarily biased the overall cash flow estimation. Despite all these limitations, this study provided supportive evidence of the potential economic benefits of implementing health monitoring programs for cows

in early lactation that rely exclusively on AHM system alerts and data rather than visual observation of clinical signs of disease to select cows for clinical examination.

CONCLUSIONS

We concluded that economic benefits can be expected when implementing health monitoring programs that rely exclusively on AHM system alerts and data rather than visual observation of clinical signs of disease to select cows for clinical examination in early lactation. The use of AHM resulted in greater milk income that fully offset and surpassed the extra costs of health monitoring and management when using an AHM system including neck-attached sensors of rumination time and physical activity and milking parlor daily milk weights. Ultimately, the results of this study demonstrated that implementing health monitoring programs that rely almost exclusively on alerts from AHM systems might result in improved herd profitability as compared with nonintensive programs that rely exclusively on visual observation of clinical signs of disease to select cows for examination.

NOTES

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Nonstandard abbreviations used: AHM = automated health monitoring; AI = artificial insemination; AMS = automated monitoring systems; AT = activity time; DA = displaced abomasum; IMM = intramammary; INDIG = indigestion; IOFC = milk income over feed costs; KET = clinical ketosis; MAST = clinical mastitis; MET = metritis; MY = milk yield; PNEU = pneumonia; VO = visual observation.

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